AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 4 · STUDENT LESSON

Rules or Learned Patterns?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Before you start

Week 1: identify human control, a written rule and learning from examples.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Execute a stated rule including its boundary. Try the recall before opening notes.

Your learning target

I can execute a stated rule including its boundary.

I can select a rule by comparing errors on **label**led examples.

I can explain why agreement with **fitting** examples does not establish performance on new cases.

Remember before reading

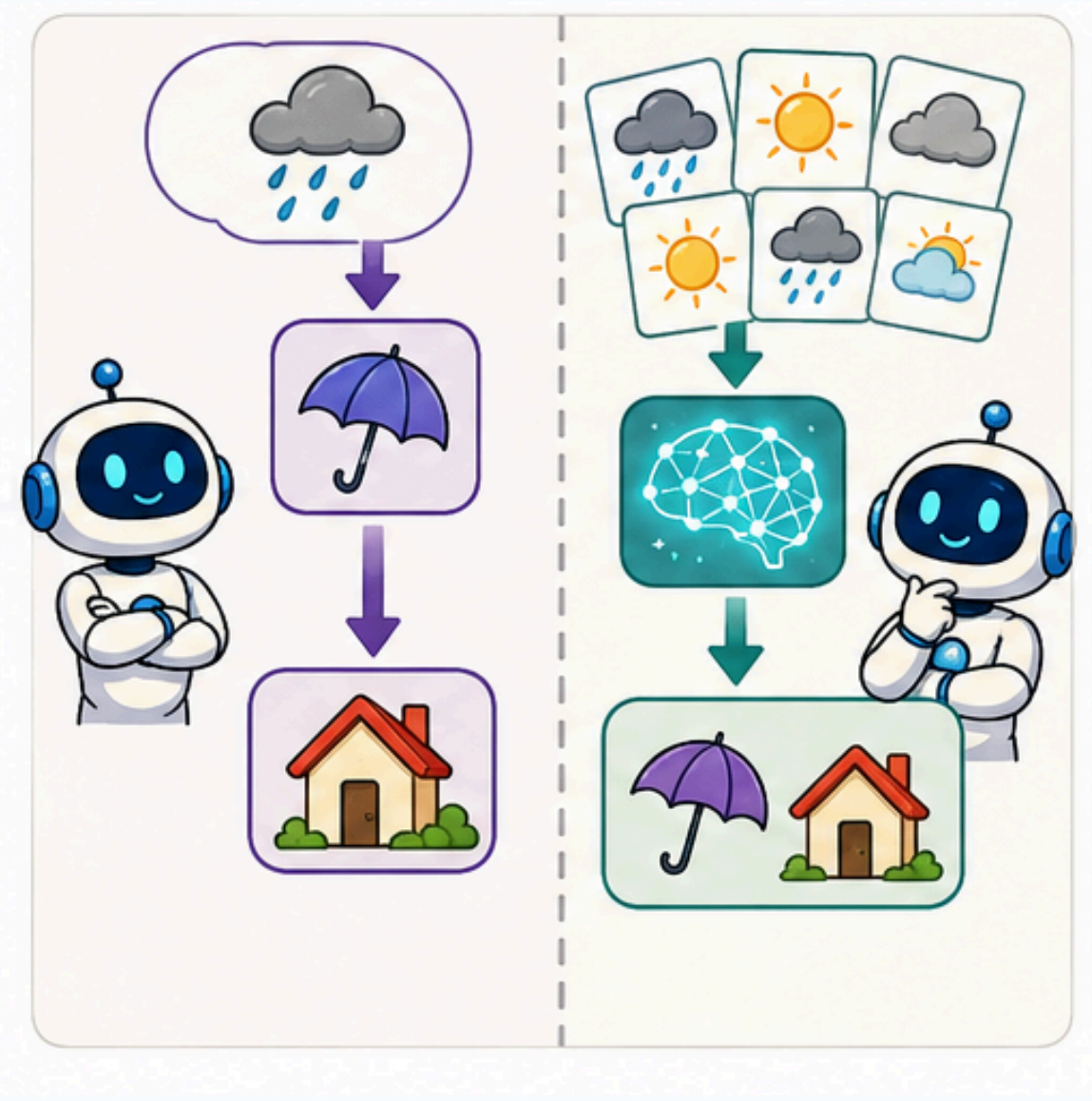
Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Sensors Are Messengers. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from AI or Not?. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Fixed instructions and learned behavior can look similar but arise differently.



Your target

LEARNING TARGET Sorter A follows 'red left, blue right'. Sorter B was trained using **label**led red and blue cards. Both send a purple card left. Does their shared **output** show that they used the same **process**?

What **evidence** is missing?

Learner name:

Readiness for the connected explanation

A warning uses temperature greater than 28 degrees C. What happens at exactly 28, and what must be checked before comparing a recorded number with 28?

This week's end goal

By the end, I can distinguish choosing a rule from applying an already chosen rule.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Rules or Learned Patterns? case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



Start with this puzzle

MYSTERY A motion light and an animal recognizer both react to **input**, but one checks a written condition while the other uses patterns learned from examples.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

A paper sorting task describes imaginary cards. Each card has a width in centimetres and a recorded **label**, narrow or wide. One person writes a rule from a design requirement: **label** a card wide when its width is at least 5 cm.

Another person compares several candidate **threshold**s with **label**led examples and chooses the one with fewest disagreements. Both may end up using the same **threshold**, but they selected it in different ways.

Executing a rule means applying it to an **input**. **Fitting** means using examples to select or adjust the rule or model. A fitted model can be very simple.

We do not need a brain-like picture or a complicated computer to study the difference between selecting a rule and using it.

In this exercise the **label**s are supplied by the task designer. They are not a universal definition of wide, and a centimetre value alone would not determine every possible meaning of that word.

State what the recorded **label**s mean in the task before counting disagreements.

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Rules or Learned Patterns?. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.



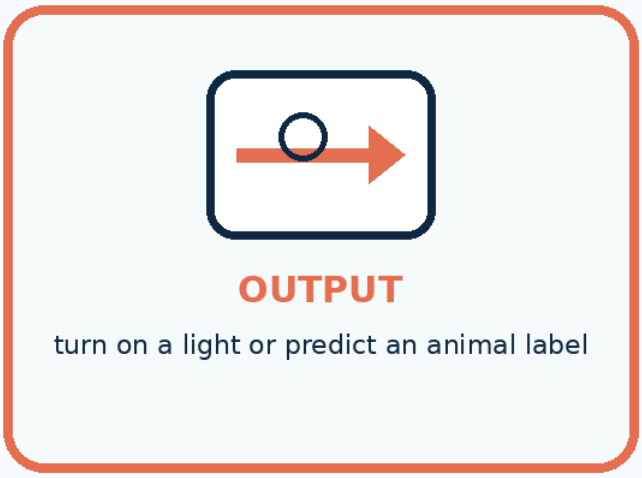
INPUT

a motion value or image values



PROCESS

run an explicit if/then rule or a learned classifier



INPUT a motion value or image values

PROCESS run an explicit if/then rule or a learned classifier

OUTPUT turn on a light or predict an animal **label**

Key idea

CORE IDEA Fixed instructions and learned behavior can look similar but arise differently.

How the mechanism works

UNDER THE HOOD A fixed program's decision logic is specified directly by people. A **learned model**'s parameters are fitted from examples.

Both are software, both may be complex, and both can fail; the distinction is how the decision behavior was produced.

Draw a fresh example with all three stages:

A closer explanation

How a threshold is selected and then applied

INFORMATION AVAILABLE	OPERATION	RESULT
Widths with supplied narrow or wide labels	Compare candidate rules by error count	Select a threshold
Selected threshold and a new width	Apply the at least comparison	Predicted label
Prediction and a later comparison label	Check for agreement	Evidence about this case

The comparison label is available during fitting and later checking. A new prediction must be made before revealing that case's label.

For **threshold** t, our rule predicts wide when width is at least t and narrow otherwise. At least includes equality: with t=5, width 5 is wide. Changing at least to greater than changes the rule and may change the **error count**.

A **fitting** procedure also contains choices people made: the **feature**, candidate **threshold**s, **label** definition and way of counting errors. Calling a **threshold** learned does not remove human responsibility for those choices.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

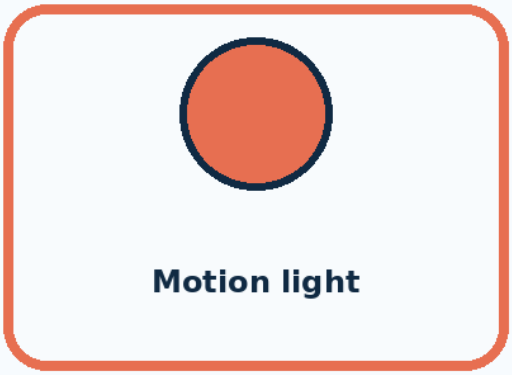
Compare the cases

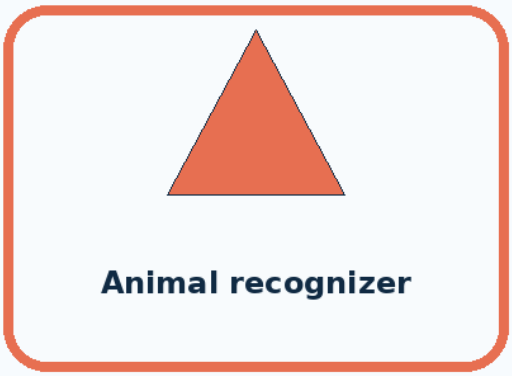
> Compare the supplied cases for Rules or Learned Patterns?. Identify the detail that changes the conclusion.

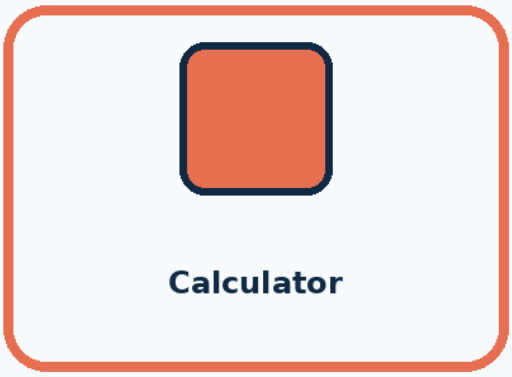
PICTURE GALLERY

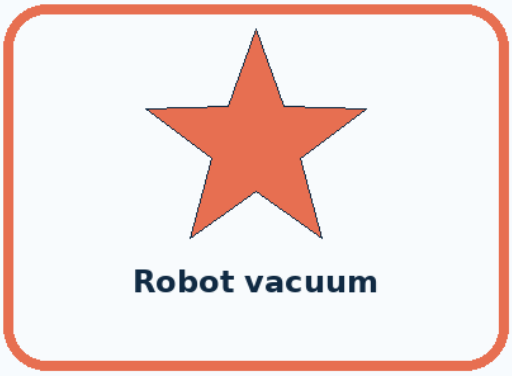
Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









The case to investigate

CASE 1 — Motion light Clue: IF motion, THEN light on. Meaning: **fixed rule**.

The case to investigate

CASE 2 — Animal recognizer Clue: trained on **label**ed images. Meaning: **learned pattern**.

The case to investigate

CASE 3 — Calculator Clue: follows arithmetic procedures. Meaning: fixed even when

complex.

The case to investigate

CASE 4 — Robot vacuum Clue: behavior alone is incomplete **evidence**. Meaning: could be fixed, learned, or mixed.

Keep this boundary clear

BOUNDARY Do not use complicated versus simple as a substitute for fixed versus learned.

Which two pictures are easiest to confuse? What **evidence** separates them?

Change one thing in the pictured case

Change one thing

Change: Give a new pebble diameter 5 instead of 4.

Predicted result

Prediction: The selected rule predicts large.

Why the result follows

Reason: At least includes equality, so 5 meets the boundary.

STUDENT LESSON · 5 OF 18

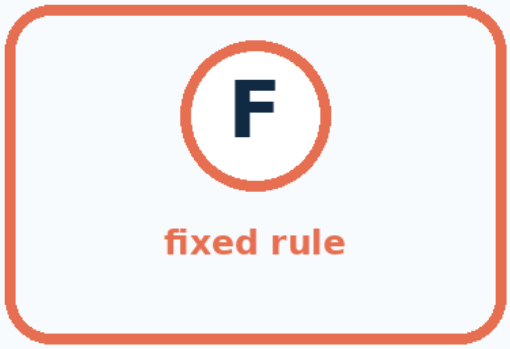
Words that unlock the idea

> Use the vocabulary of Rules or Learned Patterns? to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

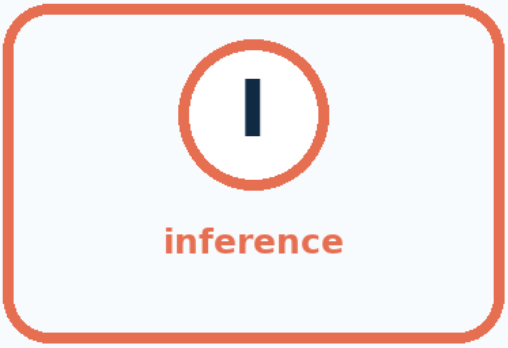
Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Word	Meaning	My example
fixed rule	instruction written directly in advance	
learned pattern	relationship fitted from examples	
training	process that adjusts a model from examples	
inference	using the finished model on a new input	

Misconception detective

Find the unsupported claim

CLAIM TO REPAIR If the **output** changes, the system must be learning.

Use one vocabulary word and one piece of **evidence** to repair it:

> WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word	Meaning
Feature	A recorded property used as an input to a decision or model
Label	A recorded category or target associated with an example
Threshold	A boundary value used in a comparison
Fitting	Choosing or adjusting a model using examples
Error count	The number of predictions that disagree with the comparison labels

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STUDENT LESSON · 6 OF 18

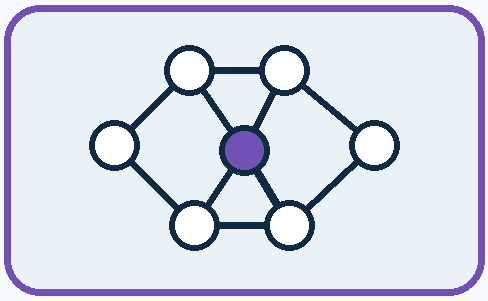
Follow a worked example

> Follow the worked trace for Rules or Learned Patterns?, predict a step, then compare with its explanation.

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



The case to investigate

CASE A motion light and an animal recognizer both react to **input**, but one checks a written condition while the other uses patterns learned from examples.

Step 1

STEP 1 — OBSERVE a motion value or image values

Step 2

STEP 2 — TRACE run an explicit if/then rule or a learned classifier

Step 3

STEP 3 — CONCLUDE turn on a light or predict an animal **label**

Model answer

MODEL REASONING The motion light follows a directly written sensor condition. The animal recogniser uses a model fitted on **label**led examples.

Observing that both react to **input** does not tell us how their decisions were produced.

A claim to avoid

DO NOT OVERCLAIM Do not use complicated versus simple as a substitute for fixed versus learned.

What new **evidence** could change this answer?

A closer explanation

Width in cm	Supplied label	Threshold 3 prediction	Threshold 5 prediction
2	narrow	narrow	narrow
4	narrow	wide	narrow
6	wide	wide	wide
8	wide	wide	wide

Worked explanation · Rules or Learned Patterns?

Apply **threshold** 3 to each width before checking the **label**. Only width 4 disagrees, so its **error count** is 1. **Threshold** 5 agrees on all four examples, so its **error count** is 0. Of these two candidates, this procedure selects 5.

Zero disagreements here means it fits these four **label**s; it is not a guarantee about all future cards.

Worked explanation · Rules or Learned Patterns?

There can be ties. **Threshold** 6 also agrees on these four examples because 6 is included by at least. **Threshold**s 5 and 6 predict differently for width 5, which is absent from the **fitting** set.

The available examples therefore do not uniquely determine how that width should be **label**led. A procedure needs an explicit tie rule, and we should record the unresolved case.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Rules or Learned Patterns?

A fictional paper sorter learns a size boundary for toy pebbles.

The facts we will use

Supplied facts

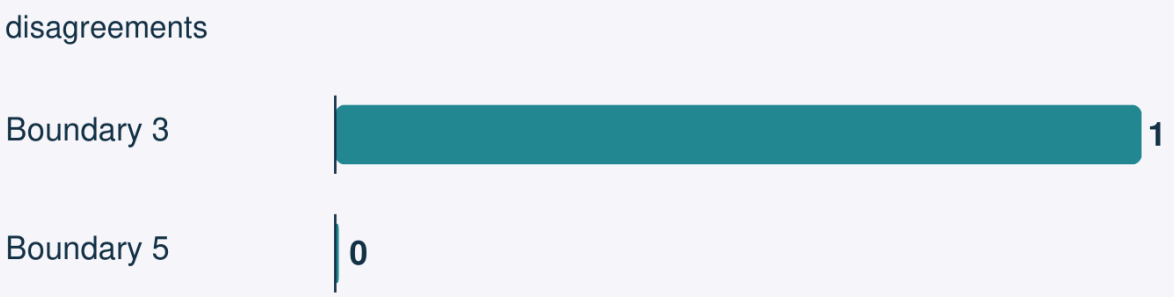
Training diameters are 1,3,5,7 cm with **label**s small,small,large,large.

Supplied facts

Compare large-at-least-3 with large-at-least-5; choose fewer errors.

AI ACADEMY / AI-1/4

Training errors by candidate boundary



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/4

Rules or Learned Patterns?

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Training examples

Diameters 1,3,5,7 have reference labels S,S,L,L.



2. Candidate 3

Predictions S,L,L,L: one disagreement at 3.



3. Candidate 5

Predictions S,S,L,L: zero disagreements.



4. New pebble

Selected boundary 5 predicts small for a new diameter 4.

CHANGE ONE THING

Give a new pebble diameter 5 instead of 4.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Training examples	Diameters 1,3,5,7 have reference labels S,S,L,L.
2. Candidate 3	Predictions S,L,L,L: one disagreement at 3.
3. Candidate 5	Predictions S,S,L,L: zero disagreements.
4. New pebble	Selected boundary 5 predicts small for a new diameter 4.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The learning procedure compares candidate rules against supplied examples and selects boundary 5. Applying that boundary to diameter 4 is a later **prediction** step.

An explicit **threshold** can be selected from data; being easy to write does not make its origin unknowable.

What this example does not establish

Keep the stated limit

Zero **training** disagreements do not prove every future pebble receives the correct **label**.

Keep the stated limit

Read the **label**led picture: Show which stage used **label**s and which stage used only the new measurement.

Keep the stated limit

Change one thing: Give a new pebble diameter 5 instead of 4. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Rules or Learned Patterns? claim and explain a repair using the supplied evidence.

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Motion light	IF motion, THEN light on	fixed rule

Case	Evidence	What it shows
Animal recognizer	trained on labeled images	learned pattern
Calculator	follows arithmetic procedures	fixed even when complex
Robot vacuum	behavior alone is incomplete evidence	could be fixed, learned, or mixed

Repair two tempting claims

Challenge claim A

CLAIM A If the **output** changes, the system must be learning.

Repair:

Challenge claim B

CLAIM B A fixed program can never be useful or intelligent-looking.

Repair:

A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A closer explanation

Check the claim against the facts

Suppose a new card has width 5 and its supplied task **label** is narrow. The selected **threshold** 5 predicts wide, so it disagrees on this case. Do not erase the example to preserve a perfect score.

> Record the error and investigate whether the **label**s follow a different definition, whether measurement matters or whether width alone is insufficient.

For this week's paper task, a **prediction** is only a proposed category. It does not authorise a physical action.

More consequential applications require additional checks and **human decision**s that a small **fitting** exercise cannot establish.

> **Explain the difference between choosing a threshold using examples and applying that threshold to a new card**

STUDENT LESSON · 8 OF 18

Practise together

> Practise Rules or Learned Patterns? with support, then explain the deciding step in your own words.

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Fixed instructions and learned behavior can look similar but arise differently.

Use a fresh example to explain **fixed rule**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: A fixed program can never be useful or intelligent-looking.

Invent one edge case and name the safe response.

Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Now use widths 1, 3, 5 and 7 with **label**s narrow, narrow, wide and wide. Compare candidates 3 and 5, using the same at least rule. Write all four **prediction**s for each candidate before you count.

This prevents a plausible-looking total from hiding a boundary mistake.

List the predictions and error count for each candidate. Select the lower-error candidate and state exactly what the result establishes

Keep this comparison separate from a final check on unseen cases. If we repeatedly choose a **threshold** after looking at the check answers, those cases have helped us fit the rule.

They no longer provide an independent check of our finished choice.

Practise with less help: Complete the supported comparison

Use workbook W2 for the connected example “Rules and **learned pattern**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Rules and **learned pattern**s. Proposed explanation: “A learned rule cannot be written simply”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Show which stage used **label**s and which stage used only the new measurement.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

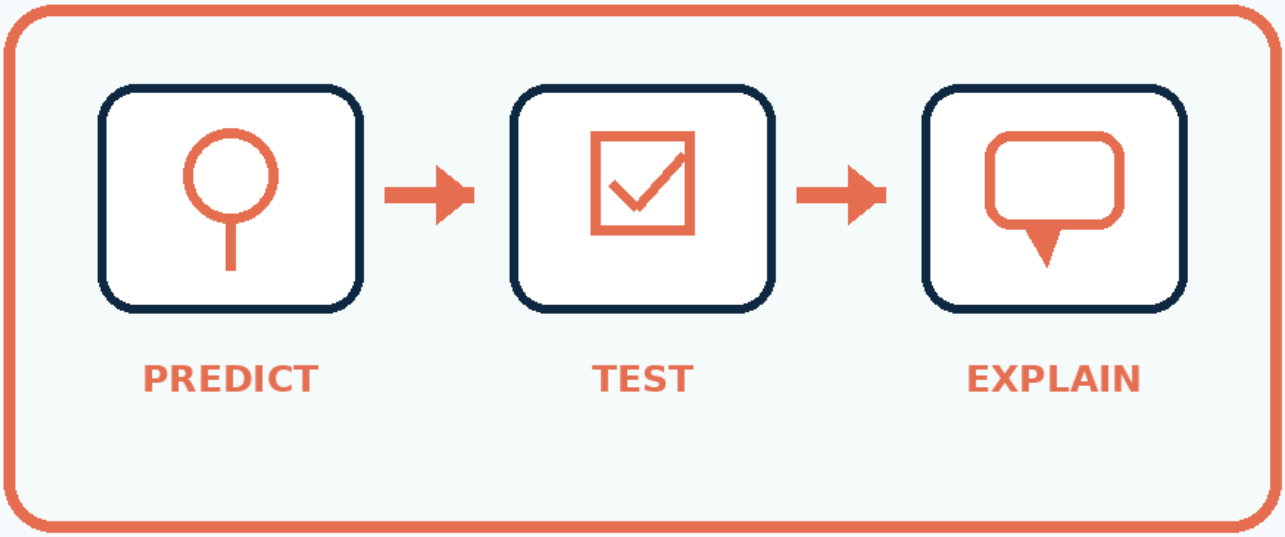
Run the investigation

> Investigate Rules or Learned Patterns?. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

Same Behavior, Different Cause

Predict first, test honestly, and explain what the **evidence** means.



1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Compare a motion light with an animal recognizer.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

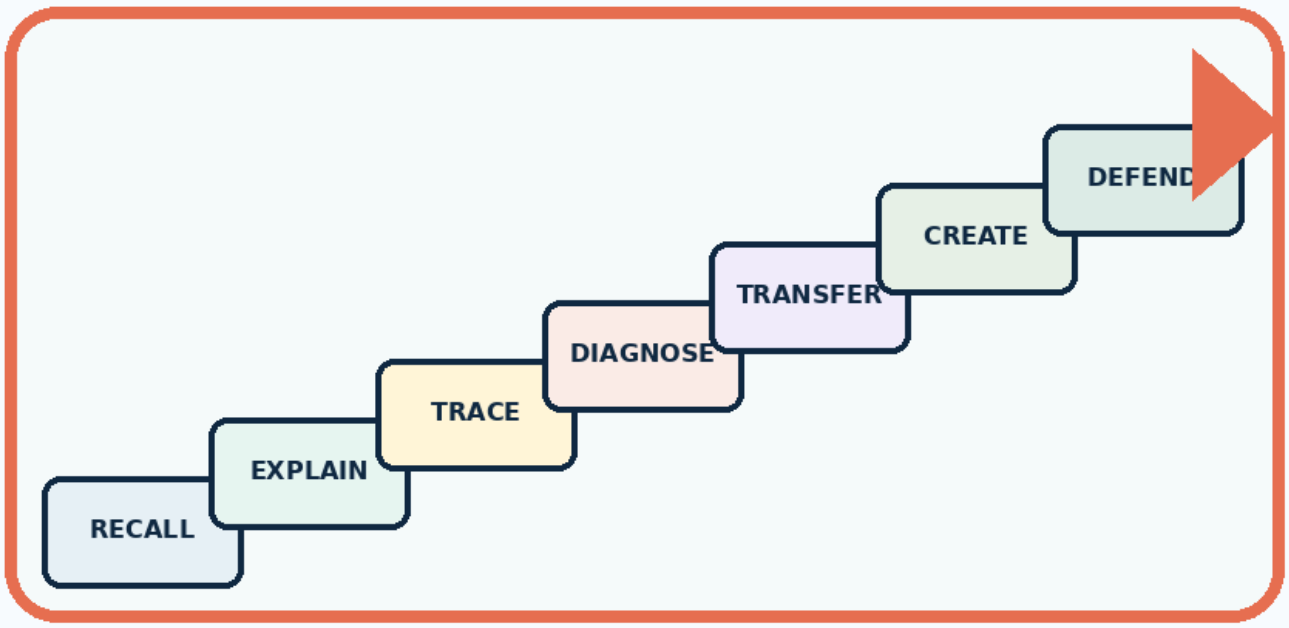
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Rules or Learned Patterns?. Explain why each step is justified.

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define **fixed rule**.

EXPLAIN Fixed instructions and learned behavior can look similar but arise differently.

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: If the **output** changes, the system must be learning.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

STUDENT LESSON · 11 OF 18

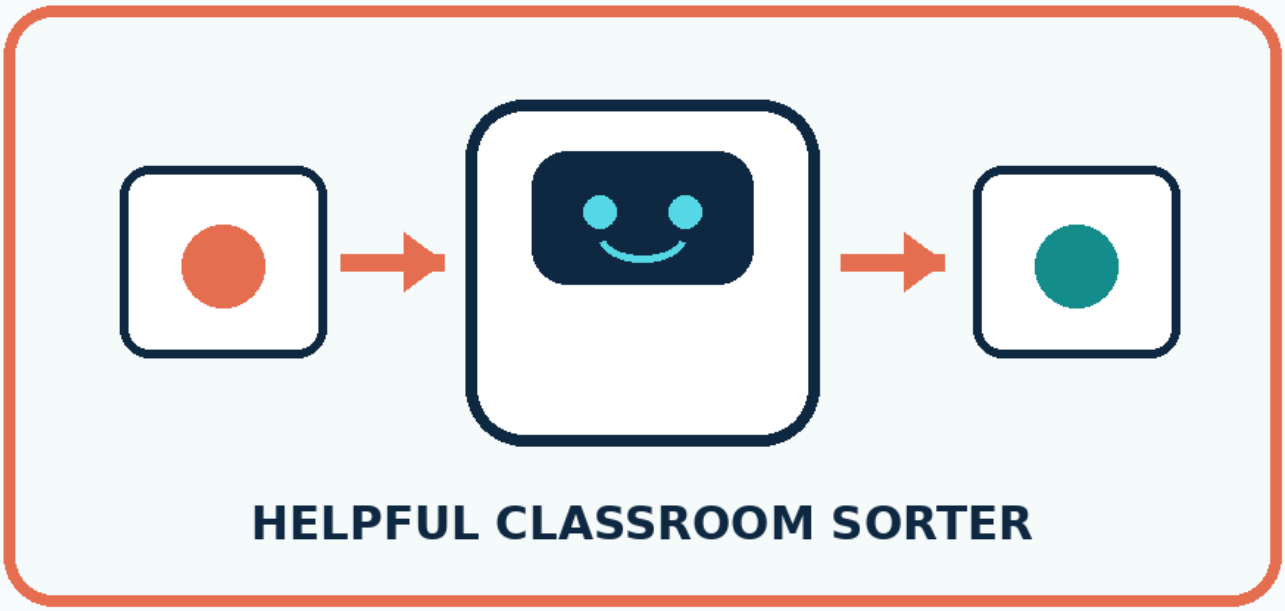
Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.



THIS WEEK'S CHECKPOINT Draw a fixed-rule version and a learned-pattern version of the Classroom Sorter; state what **evidence** would justify the learned version.

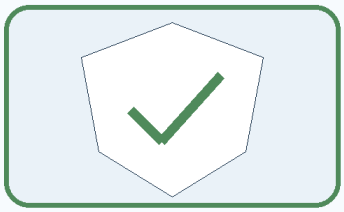
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



Responsible design

RESPONSIBLE DESIGN Do not use complicated versus simple as a substitute for fixed versus learned.

My project decision:

Evidence for your project

Document whether the sorter’s rules were teacher-written or chosen from examples, retaining candidate errors.

STUDENT LESSON · 12 OF 18

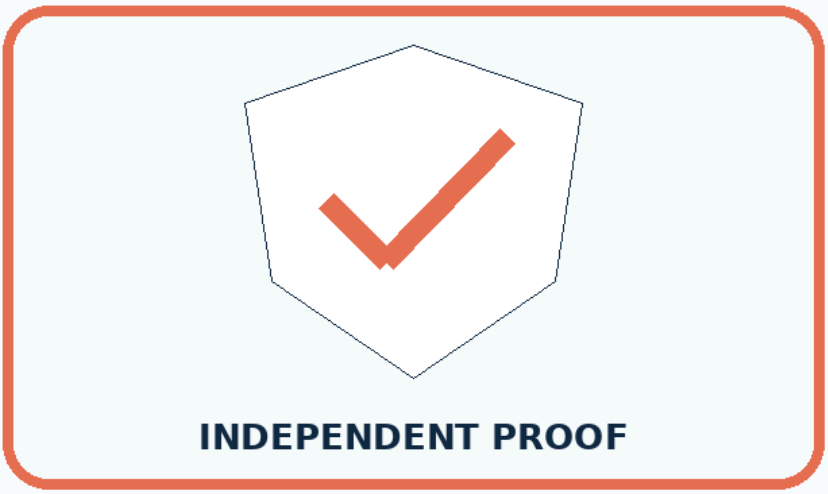
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



What success looks like

REQUIRED PROOF Two feeders release food at noon. One checks the clock; the other predicts a feeding time from past activity. Explain the **mechanism** difference.

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

Check your explanation

SELF-CHECK The motion light follows a directly written sensor condition. The animal recogniser uses a model fitted on **label**led examples. Observing that both react to **input** does not tell us how their decisions were produced.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: Pictures Become Numbers.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Rules or Learned Patterns? case for supported practice. Repeated examples do not become fresh assessments.

Week 4 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Sorter A follows a written red-card-left rule. Sorter B is fitted using **label**led leaves.

The question

How were their decision **process**es obtained? Where should A send blue cards?

Worked explanation

A was explicitly programmed; B learned parameters from examples. Blue-card handling is unspecified, so request a rule. Both systems can fail.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

Scope note: this course mainly studies learning from examples. AI also includes symbolic and rule-based approaches. Ask how a system reasons or learns; a robot body or a simple automatic switch is insufficient **evidence**.

Trace the supplied case visually

Where did the decision come from?

Evidence or stage	Value or result
Written rule	Red card goes left
Learned pattern	Fit using labelled leaves
Missing rule	Blue-card action unspecified

Sorter A follows a written red-card-left rule. Sorter B is fitted using **label**led leaves.

> How were their decision **process**es obtained? Where should A send blue cards?

A was explicitly programmed; B learned parameters from examples. Blue-card handling is unspecified, so request a rule. Both systems can fail.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Fit a new paper classifier

A fictional sorting task has lengths 2, 5, 7, 9 cm with **label**s short, short, long, long. Candidate rules predict long when length is at least 5 or at least 7.

List **prediction**s and errors for both, choose one, and predict a new length of 6 before any **label** is revealed. State one limit of your conclusion.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 4 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

System A uses the **written rule** green means go. System B learns fruit **label**s from example photographs. Who supplies A's rule? What changes when B is trained on different examples?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Draw a fixed-rule version and a learned-pattern version of the Classroom Sorter; state what **evidence** would justify the learned version.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

> Sorter A follows 'red left, blue right'. Sorter B was trained using **label**led red and blue cards. Both send a purple card left. Does their shared **output** show that they used the same **process**? What **evidence** is missing?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

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Your lesson route

> Use this map to navigate Rules or Learned Patterns?, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional paper sorter learns a size boundary for toy pebbles.

By the end, I can	How I will show it
Execute a stated rule including its boundary	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Select a rule by comparing errors on labelled examples	Explain the deciding step, show a trace or test, and state one limit.
Explain why agreement with fitting examples does not establish performance on new cases	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

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Finish and return

> Explain Rules or Learned Patterns? without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Execute a stated rule including its boundary

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional paper sorter learns a size boundary for toy pebbles.

Practice 1: explain the deciding evidence

Which **label**led step matches this detail? Selected boundary 5 predicts small for a new diameter 4.

1. **Training** examples

2. Candidate 3

3. Candidate 5

4. New pebble

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: New pebble Selected boundary 5 predicts small for a new diameter 4. The learning procedure compares candidate rules against supplied examples and selects boundary 5.

Applying that boundary to diameter 4 is a later **prediction** step. An explicit **threshold** can be selected from data; being easy to write does not make its origin unknowable.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Give a new pebble diameter 5 instead of 4. Which conclusion is supported by the supplied facts?

1. The selected rule predicts large.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The selected rule predicts large. At least includes equality, so 5 meets the boundary.

Repair a missing building block

Review Sensors Are Messengers (ai-1/3). Goal: Describe a measurement using its quantity, value, unit and conditions. Short practice: Why can we calculate the earlier gap but not predict an exact new B reading?.

Return to this week and explain what you changed.

Review AI or Not? (ai-1/1). Goal: Distinguish an observed result from an explanation of how it was produced. Short practice: Point to the learned stage. Does the changed display prove the weather forecast changed?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

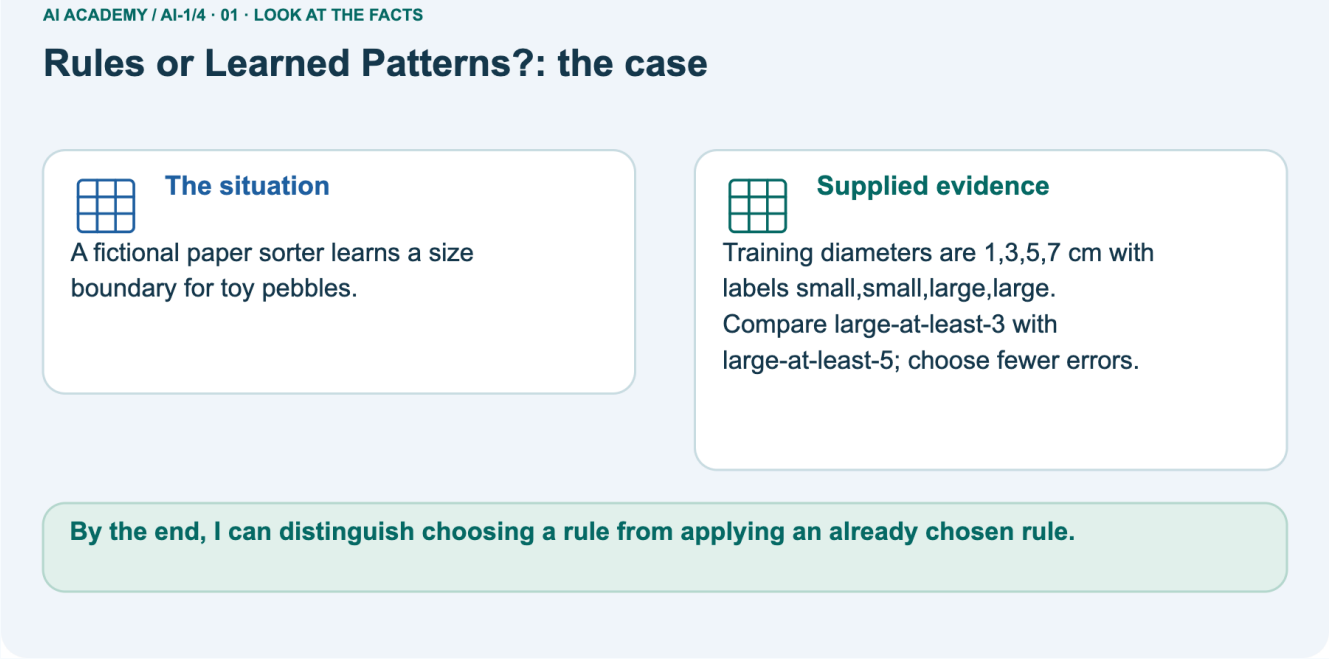
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Rules or Learned Patterns? graphic using its labels, deciding evidence and relevant limitation.

01. Rules or Learned Patterns?: the case

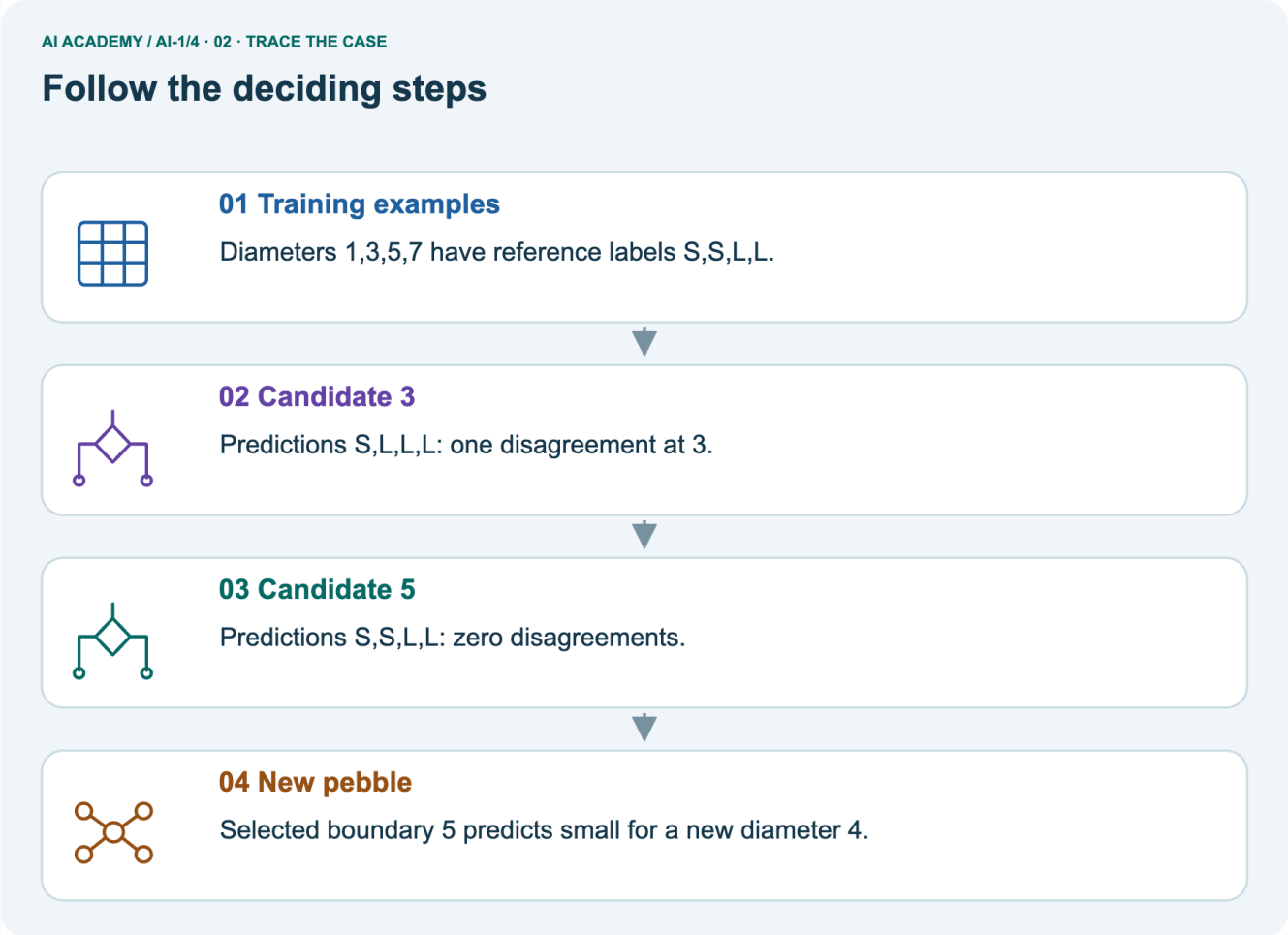


Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional paper sorter learns a size boundary for toy pebbles. Supplied facts: **Training** diameters are 1,3,5,7 cm with **label**s small,small,large,large. Compare large-at-least-3 with large-at-least-5; choose fewer errors.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

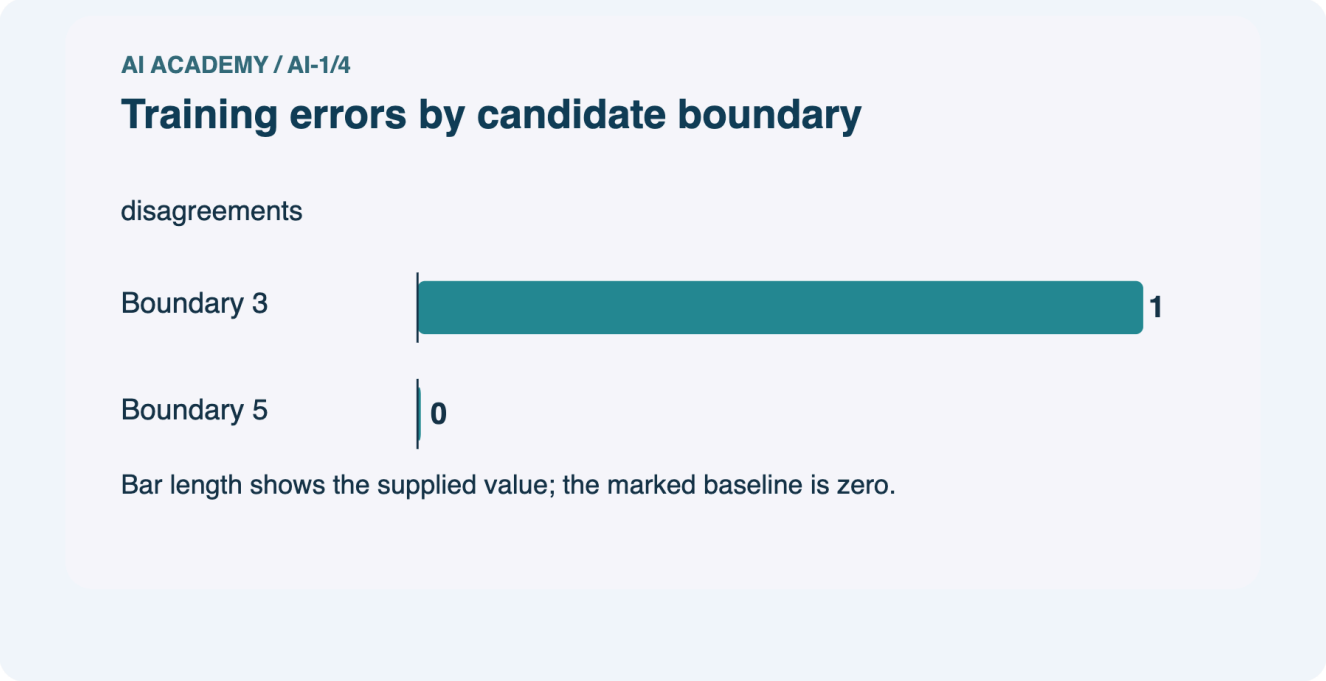


Follow the labelled picture with the matching explanation. Open the vector diagram

Training examples: Diameters 1,3,5,7 have reference **label**s S,S,L,L. → Candidate 3: **Prediction**s S,L,L,L: one disagreement at 3. → Candidate 5: **Prediction**s S,S,L,L: zero disagreements. → New pebble: Selected boundary 5 predicts small for a new diameter 4.

Point to each numbered stage and explain which recorded **input** it uses.

03. Training errors by candidate boundary

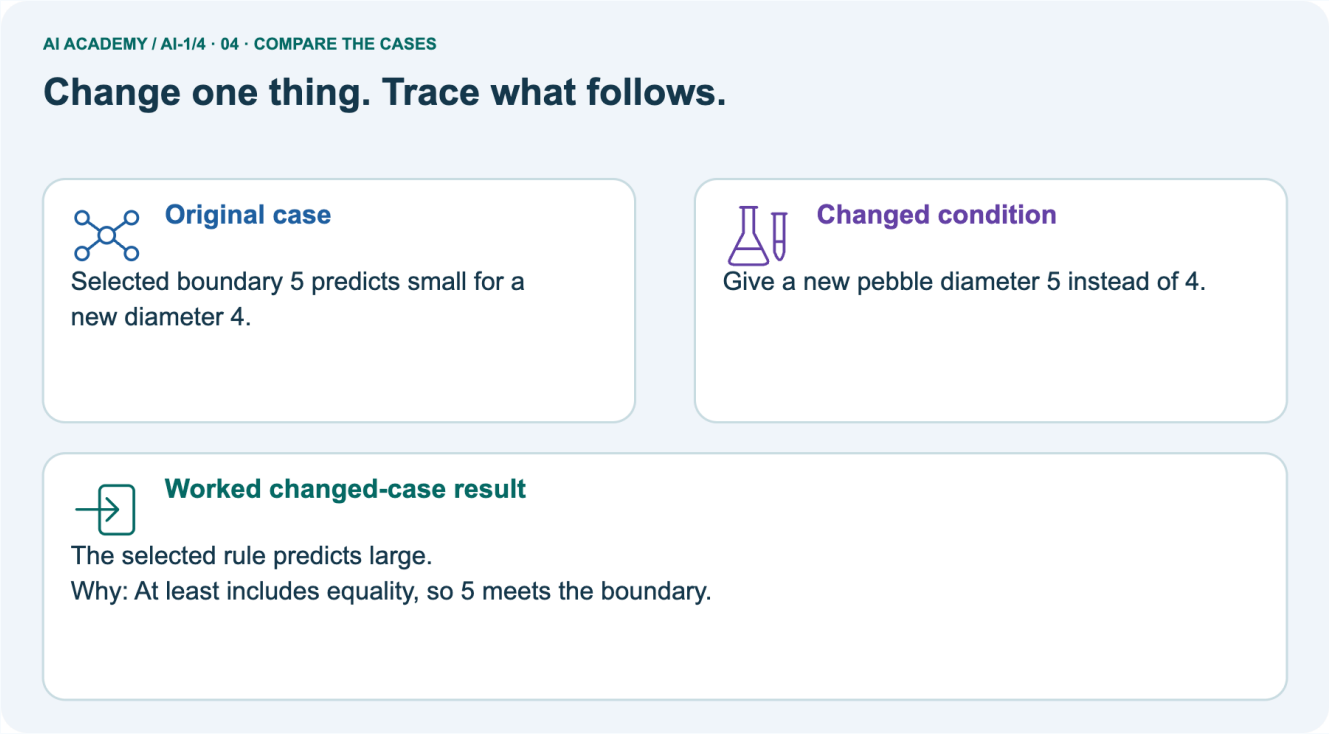


Follow the labelled picture with the matching explanation. Open the vector diagram

Training errors by candidate boundary. Units: disagreements. Boundary 3: 1. Boundary 5: 0.

Read the **label**s and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



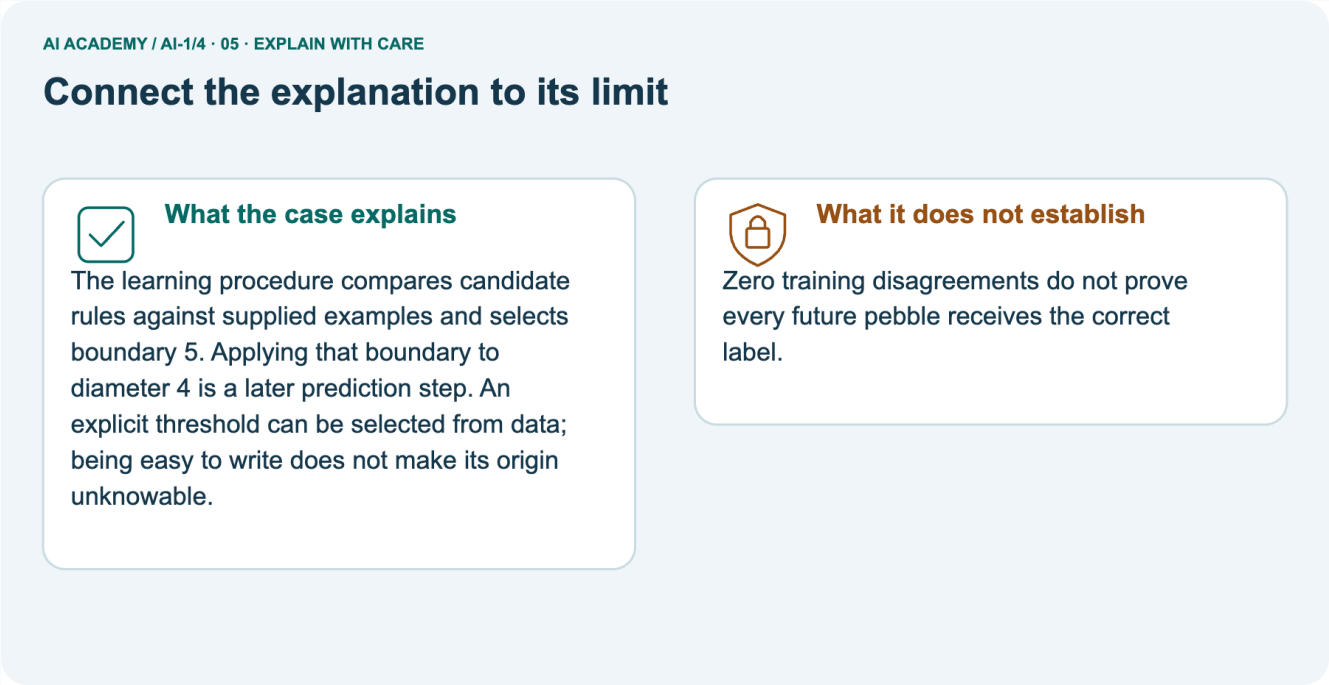
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Selected boundary 5 predicts small for a new diameter 4. Changed condition: Give a new pebble diameter 5 instead of 4. Expected result: The selected rule predicts large.

Reason: At least includes equality, so 5 meets the boundary.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



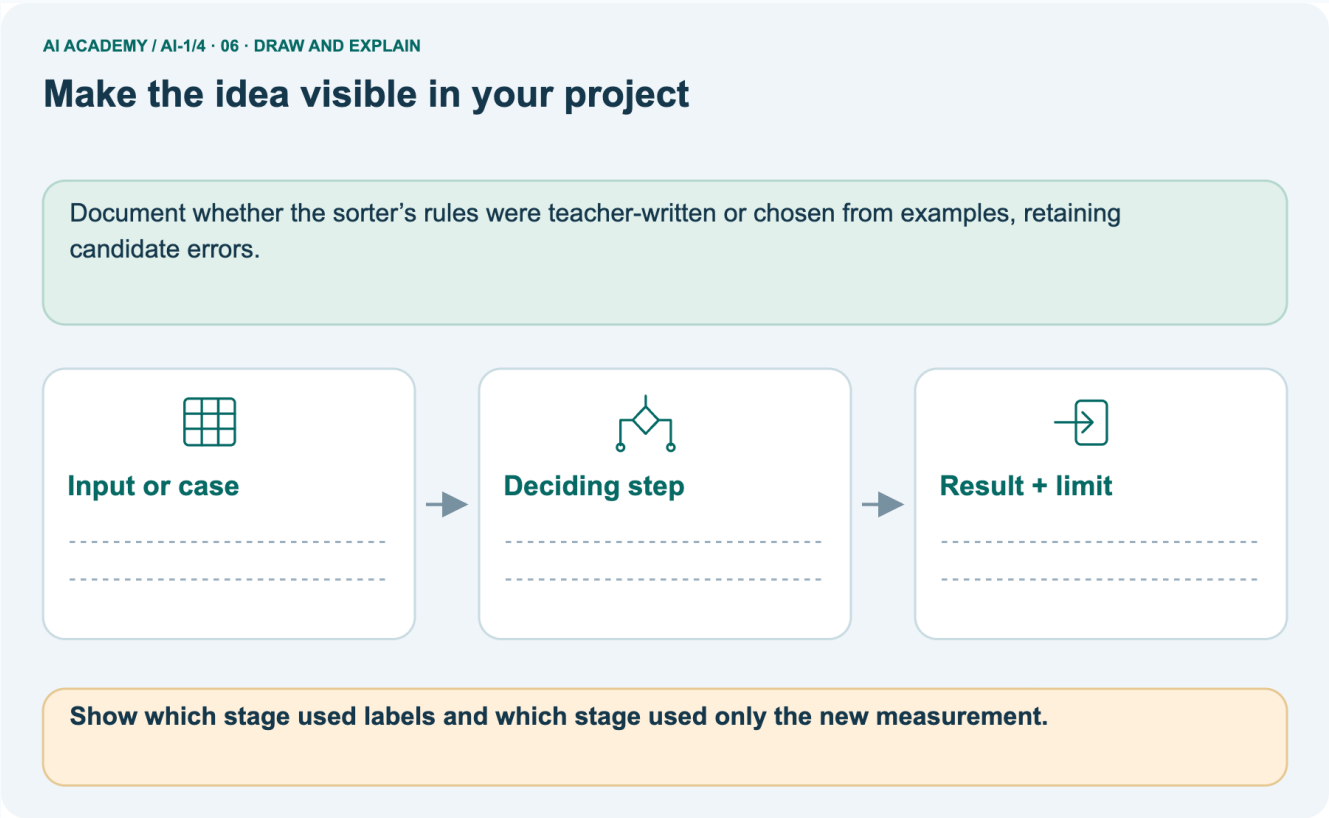
Follow the labelled picture with the matching explanation. Open the vector diagram

The learning procedure compares candidate rules against supplied examples and selects boundary 5. Applying that boundary to diameter 4 is a later **prediction** step.

An explicit **threshold** can be selected from data; being easy to write does not make its origin unknowable. Limit: Zero **training** disagreements do not prove every future pebble receives the correct **label**.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Document whether the sorter's rules were teacher-written or chosen from examples, retaining candidate errors. Draw your **input**, deciding step, result and limit.

Supported question: Show which stage used **label**s and which stage used only the new measurement.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.